ABHAY SINGH Data Analyst

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PROFESSIONAL SUMMARY

Results-driven Data Science professional with experience in machine learning, data analytics, and statistical modeling. Proficient in Python, SQL, and Power BI with a proven track record of building ML-powered applications and delivering actionable insights from complex datasets.

EDUCATION

Divine Public School, Gorakhpur, Uttar Pradesh
10th and 12th

SR Institute of Management and Technology, Lucknow, Uttar Pradesh
B.Tech in Computer Science

SKILLS

Programming Languages
- Python Programming
- SQL
- C Language

Data Analysis & Visualization
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Exploratory Data Analysis (EDA)
- Data Visualization
- Jupyter Notebook

Database
- MySQL
- Database Design
- Query Optimization

Deep Learning
- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- TensorFlow & Keras

Machine Learning
- Machine Learning
- scikit-learn
- Predictive Modeling
- Statistical Analysis

Data Engineering & Tools
- Data Cleaning
- Data Preprocessing
- Data Validation
- ETL Pipelines
- Excel (Advanced Functions)
- Statistics

Web Development
- HTML
- CSS
- JavaScript
- Bootstrap
- Django

EXPERIENCE

Softpro India 04/2025 – present | Lucknow, India
Data Science Internship
- Engineered ETL pipelines to transform large-scale datasets, improving data processing efficiency by **30%.**
- Developed predictive models using **scikit-learn** to support business intelligence and strategic decision-making.
- Created interactive **Power BI** dashboards, enabling stakeholders to visualize key metrics and trends.

- Implemented data validation frameworks to ensure data integrity across 10+ disparate data sources.

Unified Mentor 08/2024 – 10/2024 | Remote

Data Science Intern

- Analyzed **50,000+ records** using Pandas and NumPy to extract actionable business insights.
- Built and deployed ML models achieving **85%+ accuracy** for classification and predictive analytics tasks.
- Automated data cleaning workflows, reducing manual effort by **40%** and enhancing overall data quality.
- Conducted **Exploratory Data Analysis (EDA)** to identify patterns, outliers, and correlations.

CERTIFICATES

- MS Excel Online Training Course
- Learn SQL- Structured Query Language
- Python Data Science Course With Numpy, Pandas and Matplotlib
- Comprehensive Data Visualization Course: Matplotlib in Python

PROJECTS

Django Chatbot Web Application | Python, Django, Scikit-Learn, NLP

- **Chatbot Development**: Developed an chatbot web application using **Python**, **Django**, and **Scikit-Learn** forautomated user interaction and response generation.
- **NLP & Feature Engineering**: Implemented tokenization, stop-word removal, and **TF-IDF vectorization** fortext preprocessing and intent prediction.
- **Machine Learning Models**: Built and optimized **Intent**, **Category**, and **Flag** classification models using **Scikit-Learn pipelines** and **LinearSVC**.
- **Model Integration & Deployment**: Integrated trained **ML** models into the **Django** backend using **Pickle** for real-time chatbot inference and response handling.
- **Frontend & Session Management**: Designed responsive UI using **HTML**, **CSS**, **Bootstrap**, and **Django** Templates with session-based chat management.

Entertainer Career Analysis | Python, Pandas, Matplotlib

- **Multi-Source Data Integration**: Consolidated and cleaned multiple Excel-based datasets (Basic,Breakthrough, and Last Work Info) using **Pandas** to create a unified view of entertainer career lifecycles.
- **Exploratory Data Analysis (EDA)**: Performed comprehensive EDA to identify career breakthrough patterns,performance distributions, and achievement trends using **Python**.
- **Statistical Visualization**: Developed categorical and descriptive visualizations using **Matplotlib and Seaborn** to communicate complex career metrics to non-technical stakeholders.
- **Insight Synthesis**: Authored a technical summary report documenting actionable insights regardingentertainer attributes and longevity.

Car Marketplace & Management System | Python, Django, Machine Learning

- **Full-Stack Web Application Development:** Developed a full-stack car marketplace platform for buying, selling, renting, and servicing vehicles using **Python, Django, HTML, CSS, Bootstrap,** and **JavaScript.**
- **Machine Learning-Based Car Price Prediction:** Built a **Machine Learning**-based car price **prediction system** using **Scikit-Learn** and integrated it into the Django application.
- **AI-Powered Car Recommendation Engine:** Designed an AI-powered car **recommendation engine** using **Deep Learning embeddings** and **Cosine Similarity** for personalized vehicle suggestions.
- **Multi-Role Management System:** Implemented ro**le-based authentication** and dashboards for Customers, Staff, and Admins with order, service, and request management features.
- **Data Processing & Feature Engineering:** Managed **data preprocessing, feature engineering,** and **model integration** using **Pandas, NumPy, Joblib,** and **Hugging Face** model hosting.
- **Production Deployment & DevOps**:Deployed the application on **Render** with **PostgreSQL**, **Gunicorn**, **WhiteNoise**, and **cloud-hosted ML** models.

Netflix Content Analysis & Visualization | Python, Pandas, Matplotlib

- **Engineered Data Pipelines**: Developed a Python-based workflow using **Pandas and NumPy** to clean andpreprocess raw Netflix datasets, resolving missing values and duplicate records for data integrity.
- **Exploratory Data Analysis (EDA)**: Executed comprehensive EDA to identify distribution patterns betweenMovies and TV Shows, uncovering trends in release timelines and regional content production.
- **Visual Insight Generation**: Leveraged **Seaborn** and **Matplotlib** to build visualizations for genre trends andcountry-wise content saturation, summarizing findings into an actionable insights report.

Netflix AI Recommendation System | Python, Django, Scikit-Learn, Pandas, Machine Learning

- **Recommendation System Development**: Developed an AI-powered movie recommendation web application using Python, **Django**, and **Scikit-Learn** for personalized Netflix-style movie suggestions.
- **Deep Learning & Similarity Modeling**: Implemented **Deep Learning Movies Encoder** and **Cosine Similarity** algorithms to generate intelligent content-based movie recommendations.
- **Data Processing & Feature Engineering**: Performed Netflix dataset **cleaning**, **preprocessing**, **text normalization**, and **feature extraction** using **Pandas** and **NumPy**.
- **Model Integration & Deployment**: Integrated serialized **DL** recommendation models into Django backend using **Joblib** with automated model downloading from **Hugging Face** during deployment.
- **Frontend & Database Management**: Designed responsive Netflix-inspired UI using **HTML**, **CSS**, **Bootstrap**, and **JavaScript** with **PostgreSQL/ SQLlite** database integration and recommendation history tracking.
- **Cloud Hosting & Production Deployment**: Deployed the complete **Django** application on **Render** with **PostgreSQL**, **WhiteNoise static** handling, **Gunicorn** server configuration, and **cloud-hosted DL model** support.